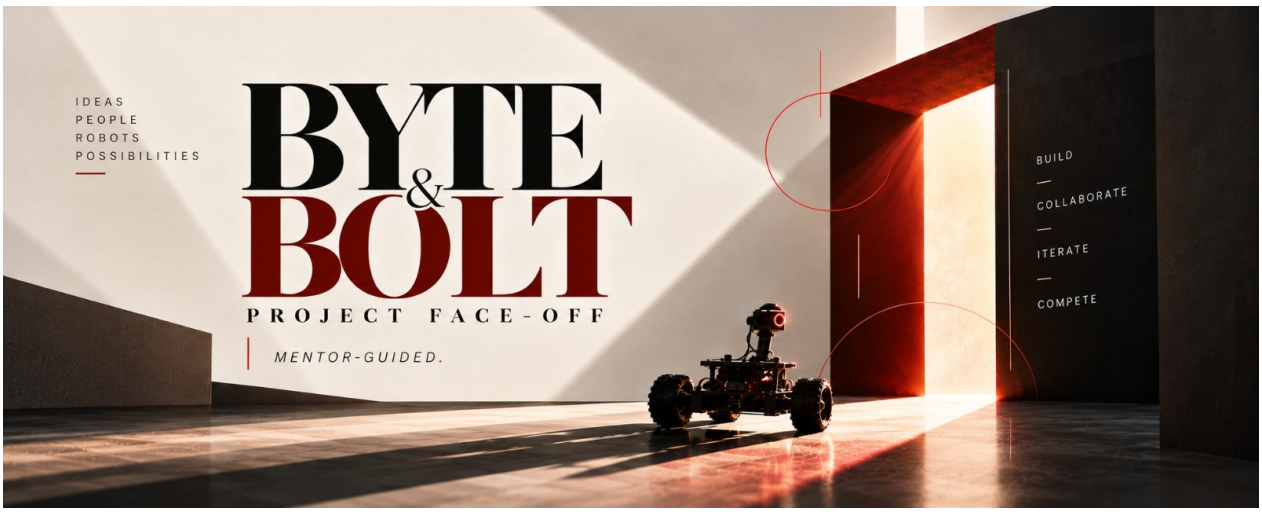


Byte and Bolt — Official Rulebook

Robotics Club | Byte and Bolt | Build • Test • Improve • Compete

Read the whole rulebook before you start. Every rule here is mandatory. This rulebook is shared online, and all updates will be posted on GitHub by the Event Head (see Section 11).

1. About the Event

Byte and Bolt is a build event run by the Robotics Club. Every team builds a working combat robot prototype.

Item	Rule
Duration	Starts 14 October, ends 4 November
Robot size	Maximum 50 cm × 50 cm × 50 cm (length × width × height)
Robot weight	Maximum 5 kg, with battery and weapon fitted
Budget	₹2,200 per team. No overspending allowed
Team	1 Team Leader, team members and 1 Mentor
Reporting	Weekly checklist, plus a progress meeting twice a week (no daily reports)

2. Who Does What

Mentor

- Guides the team on mechanical, electrical and CAD work.
- Watches the tests.
- Reviews the team once a week and signs the weekly checklist.
- Tells the Event Head immediately if there is any safety problem.
- Can suggest a resource for the team, but cannot approve it. Only the Event Head can approve.

Team Leader

- Runs the team and gives tasks to every member.
- Keeps the expense sheet and the bills (see Sections 6 and 7).

- Fills the weekly checklist honestly.
- Asks the Event Head for permission when the team needs something extra.

Team Members

- Do the work given to them.
- Follow the safety rules.
- Tell the Team Leader about any problem or failure.
- Keep the work area clean.

The Team Leader and the Mentor both report to the Event Head. Team members report to the Team Leader.

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3. Weekly Plan

Week	What the team must finish
Week 1	Design is final. Parts list (BOM) is approved by the Event Head.
Week 2	Robot is built. Wiring is done. First controlled drive works.
Week 3	Testing with the Mentor present, safety inspection, final fixes, final inspection, demo matches and judging.

If your team is behind, say so in the weekly checklist.

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4. Weekly Checklist

- The Team Leader fills one printed checklist every week. The checklist is given separately.
- The Mentor reviews the team that week, checks the checklist and signs it.
- Submit it to the Event Head at the end of the week, together with that week's stamped bills and the verified expense sheet.
- Write only what is really done. Planned work is not completed work.
- Do not sign anything that is not true.

Do not wait for the weekly checklist if any of these happen. Tell the Event Head immediately:

- a safety problem
- a damaged or swollen battery
- heat, smoke or fire risk
- the robot moving out of control
- major damage to the robot
- someone using something without permission
- a problem that affects other teams

If you are stuck, first try to solve it in your team. Then ask your Mentor. If it is still not solved, tell the Event Head.

Progress Meeting (Twice a Week)

- Every team must report its progress to the Event Head in a meeting two times every week.
- The Team Leader attends and gives the report. The Mentor should attend when possible.
- In the report, tell: what is done since the last meeting, what is planned next, what problems you have, and how much money is spent so far.

- The days and time of the meetings will be set by the Event Head and posted on GitHub (see Section 11).
- These meetings do not replace the weekly checklist. The weekly checklist is still required.

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5. Permission for Resources

Take permission from the Event Head before you use anything that was not given to your team. This includes:

- another team's parts
- spare parts, batteries, chargers, motors and electronics that are not yours
- tools and machines (3D printer, laser cutter, CNC, soldering station)
- the test area, the arena and storage areas
- extra workspace

If something is lying there and nobody is using it, it is still not yours to use.

How to ask: The Team Leader sends a message to the Event Head with:

1. Team name
2. What you need
3. Where it is
4. Why you need it
5. For how long
6. Who will be responsible

The Event Head replies: Approved, Not approved, or Approved with conditions. In an urgent case the Event Head can say yes verbally. The Team Leader then writes it in the weekly checklist.

Other rules

- Never touch another team's property without the Event Head's approval and that team's knowledge.
- Return shared things clean and in the same or better condition.
- If anything is damaged or lost, report it immediately.
- If you break this rule, your access can be stopped, you may have to replace the item, or your team can be removed from the event.

Workspace: Room 308, GICT

Room 308 (GICT) is the work room for this event. Follow these rules strictly:

- **Working time:** teams can work in Room 308 from 6:00 AM to 12:00 AM (midnight).
- **Room must be empty from 3:30 PM to 5:30 PM every day.** No team member may stay in the room during this time. Leave the room before 3:30 PM and come back only after 5:30 PM.
- **Clean before you leave.** Every team must clean its own mess before leaving the room, and especially before 3:30 PM. This means: remove scraps, packaging, tools, wires, cut material and dust; put tools and parts back in their place; and leave the tables and floor clean.
- **Leave nothing lying around.** Take your robot, parts, batteries and tools with you or store them in the approved storage place. Batteries must never be left charging alone (see Section 8).
- If a team does not clean up or stays in the room during the empty time, the Event Head can stop that team's access to the room (see the consequences under Other rules above).

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6. Budget and Expense Sheet

Budget rules

- Each team gets ₹2,200. This is the limit. No overspending, strictly.
- In Week 1, submit your parts list (BOM) to the Event Head for approval.
- Club lab items can be used only with the Event Head's permission.

Expense sheet Every team must make one Excel sheet and note every purchase in it. Keep it updated all the time. Use these columns:

Column	What to write
Date	Date of the bill
Bill No.	Bill number
GST	GST shown on the bill
Material	What you bought
Cost	Cost of the material
Quantity	How many you bought
Total	Amount for that item

At the end of the sheet, add one row with the grand total of all your spending. Before you submit, the Mentor and the Team Leader must both check and verify the sheet.

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7. How to Submit and Stamp Bills

Follow every step. No step can be skipped.

1. **Bill name:** Every bill must be made in the name of Ahmedabad University. Check this before you pay.
2. **Gate stamp:** Every bill must be stamped at the gate.
3. **Expense sheet:** Enter every bill in your Excel sheet.
4. **Verify:** Every week, the Mentor and the Team Leader check and verify the sheet before it is submitted.
5. **Weekly submission:** At the end of every week, the Team Leader submits that week's stamped bills and the verified expense sheet to the Event Head, along with the weekly checklist.
6. **Signature:** On 4 November, the Event Head will give all the collected bills to Keyur Sir for his signature.
7. **Print copy:** After the signature, each team must submit a printed copy of those bills to the respective gates.

Strict and mandatory. If any step is missed, for example the bill is not in the name of Ahmedabad University or is not stamped at the gate, we will not reimburse even a single rupee for that bill. We are not responsible if a team does not follow these steps.

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8. Robot Rules

The Event Head will confirm the exact limits (magnet, weapon speed) before the event starts and post them on GitHub.

Size and weight

- Maximum size is **50 cm × 50 cm × 50 cm** (length × width × height).
- Maximum weight is 5 kg, measured with battery and weapon fitted.

Materials

- **Robot body, frame and chassis:** 3D-printed plastic (PLA, PETG, ABS), acrylic, plywood or MDF only.
- **Metal:** Metal may be used **only for attack parts**, such as wedges, weapon tips, cutter blades and weapon bars or discs. Metal is not allowed in the body, frame or chassis.
- **Not allowed:** liquids, gels, flamethrowers or any fire or heat weapons, explosives, pressurised gas, projectiles, nets or tethers, strong magnets, and anything that can damage the arena.

Electronics and power

- Use only LiPo or Li-ion batteries.
- Charge and store LiPo batteries in a fireproof bag, and never leave them charging alone.
- The robot needs a main power switch that can be reached from outside, and a separate weapon kill switch.
- Failsafe is a must: all motors, including the weapon, must stop if the control signal is lost.
- Use a dedicated radio receiver. Shared Wi-Fi or Bluetooth is not allowed unless the Event Head approves.
- Wiring must be insulated and tied down. Battery and electronics must be fixed firmly.

Weapons

- Allowed: wedges, lifters, flippers, pushers, cutters, and low-energy drum or bar spinners (cutter motors included).
- Flamethrowers and any other fire or heat weapons are strictly not allowed.
- A spinning or cutting weapon needs a full guard and must stay under the speed limit set by the Event Head.
- The weapon must stay locked or off in the pit area and anywhere outside the arena or test box.
- Any weapon needs approval from the Mentor and the Event Head.

Matches

- A match is 2 minutes.
- A match ends on knockout, 10 seconds without moving, or leaving the arena.
- If there is no knockout, judges give points for aggression, control and damage.
- Every robot is inspected before every match (switches, wiring, battery, weapon guard).
- Judges can disqualify any unsafe robot.

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9. Safety Rules

1. Switch off power before any mechanical or electrical work, unless the Mentor has approved a test with power.
2. Follow the approved battery procedure for handling and charging. Keep damaged or swollen batteries away and report them.
3. The first power-up and first drive must be done with the Mentor present.
4. For the first motor check, keep the wheels off the ground.
5. Never test a robot that is not under control near people.
6. Wear safety glasses in the workshop and pit area, and other safety gear for fabrication.
7. Never work alone on machines or in restricted areas.

10. Scoring and Final Check

Before the final demo, the Event Head must see:

- the robot fully assembled and wiring inspected
- battery and electronics fixed securely
- the robot driving in control, tested more than once
- changes written down
- spare parts ready
- driver and pit roles decided
- the inspection checklist completed
- no safety problem pending

What is checked	Marks
Working drive and control	25
Weapon or attack mechanism	20
Design and build quality	20
Documentation, expense sheet and weekly checklists	15
Match performance	20
Total	**100**

11. Updates on GitHub

The Event Head will post all updates on GitHub. This includes rule changes, announcements, dates, and the final magnet and weapon limits. The link will be shared by the Event Head. Check it every week. Anything posted there is part of the rules.

12. Agreement

By taking part in Byte and Bolt, every Mentor, Team Leader and Team Member agrees to follow this rulebook. The Event Head's decision is final.